

# Mohamed Amine Merzouk

POSTDOCTORAL RESEARCHER · MILA & MCGILL UNIVERSITY

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## BIOGRAPHY

AI safety researcher working on the alignment of language models. I am a postdoctoral researcher at McGill University and Mila with Pr. Adam Oberman, where I study alignment methods that act on a model's internal representations, what language models encode about their own state and outputs, and honesty and deception. I come from a cybersecurity background: my PhD thesis at Polytechnique Montréal, co-supervised by Pr. Frédéric Cuppens and Reda Yaich (IRT SystemX), addressed adversarial attacks and defenses for machine learning-based network intrusion detection and was nominated for the best thesis award.

## EDUCATION

### Philosophiæ doctor (PhD) in Computer Science

Polytechnique Montréal

Montréal, Canada

January 2021 - September 2024

Thesis: *A Landscape of Adversarial Threats to Machine Learning-Based Intrusion Detection*, nominated for the best thesis award. Supervisors: Pr. Frédéric Cuppens (Polytechnique Montréal) and Reda Yaich (IRT SystemX).

### Master's degree in Computer Science

École Nationale Supérieure d'Informatique (ESI ex.INI)

Algiers, Algeria

September 2018 - September 2020

Thesis: *Adversarial Machine Learning against Anomaly-based Intrusion Detection Systems*. Supervisor: Pr. Lydia Bouzar.

### Engineer's degree in Computer Science

École Nationale Supérieure d'Informatique (ESI ex.INI)

Algiers, Algeria

September 2017 - September 2020

### Preparatory class in Mathematics and Computer Science

École supérieure en informatique 08 Mai 1945

Sidi Bel Abbès, Algeria

September 2015 - September 2017

## PROFESSIONAL EXPERIENCE

### Postdoctoral researcher

McGill University & Mila - Quebec Artificial Intelligence Institute

Montréal, Canada

October 2024 - Present

AI safety and alignment of language models, with Pr. Adam Oberman.

### Lecturer

Polytechnique Montréal

Montréal, Canada

January 2021 - August 2026

Undergraduate and graduate cybersecurity courses.

### Affiliated Doctoral researcher

IRT SystemX

Montréal, Canada

January 2021 - September 2024

Doctoral research on adversarial machine learning for network intrusion detection.

### Research intern

IMT Atlantique

Rennes, France

March 2020 - September 2020

Research internship leading to the paper *A Deeper Analysis of Adversarial Examples in Intrusion Detection* (CRISIS 2020).

### Cybersecurity audit intern

KPMG

Algiers, Algeria

July 2019 - August 2019

Implementation of the ISO 27001 information security management standard: risk assessment and drafting of security policies.

### Virtualization and cloud intern

ADEX Technology

Algiers, Algeria

June 2018 - November 2018

Design and deployment of a Red Hat OpenStack private cloud: architecture of the compute and security hardening.

### IT support

British Petroleum

Algiers, Algeria

September 2018 - October 2018

User support and workstation maintenance: troubleshooting, software installation and account management.

## TEACHING EXPERIENCE AT POLYTECHNIQUE MONTRÉAL

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- INF8102** **Security in Cloud Environments.** Lecturer: Fall 2025.
- INF8085** **Cybersecurity.** Lecturer: Summer 2025, Winter 2026, Summer 2026.
- CR345** **Security of Servers.** Lecturer: Winter 2025.
- INF4420** **Introduction to Computer Security.** Lecturer: Summer 2023, Summer 2024. Lab instructor: Winter 2022, Fall 2022, Summer 2023, Summer 2024.
- INF6103** **Cybersecurity of Critical Infrastructures.** Lab instructor: Winter 2022, Fall 2022, Fall 2023, Fall 2024.
- INF8602** **Operating Systems Security.** Lab instructor: Winter 2022.
- INF1040** **Introduction to Computer Engineering.** Evaluator: Winter 2021, Fall 2021.

## PUBLICATIONS

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### JOURNAL ARTICLES

Efficiency vs. Alignment: Investigating Safety and Fairness Risks in Parameter-Efficient Fine-Tuning of LLMs

Mina Taraghi, Yann Pequignot, Amin Nikanjam, Mohamed Amine Merzouk, Foutse Khomh

*IEEE Transactions on Artificial Intelligence* (2026). 2026

Adversarial robustness of deep reinforcement learning-based intrusion detection

Mohamed Amine Merzouk, Christopher Neal, Joséphine Delas, Reda Yaich, Nora Boulahia-Cuppens, Frédéric Cuppens

*International Journal of Information Security* (2024). Springer, 2024

Investigating the practicality of adversarial evasion attacks on network intrusion detection

Mohamed Amine Merzouk, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich

*Annals of Telecommunications* 77.11 (2022) pp. 763–775. Springer, 2022

### CONFERENCE PROCEEDINGS

How Much is Left? LLMs Linearly Encode Their Remaining Output Length

Mohamed Amine Merzouk, Dmitri Carpov, Mirko Bronzi, Damiano Fornasiere, Adam Oberman

*Advances in Neural Information Processing Systems (NeurIPS)*, 2026

Efficient Safety Alignment of Language Models via Latent Personality Traits

Mohamed Amine Merzouk, Nolan Smyth, Damiano Fornasiere, Linh Le, David Williams-King, Adam Oberman

*Conference on Language Modeling (COLM)*, 2026

Latent Personality Alignment: Improving Harmlessness Without Mentioning Harms

Linh Le, David Williams-King, Mohamed Amine Merzouk, Aton Kamanda, Adam Oberman

*Trustworthy AI Workshop, International Conference on Learning Representations (ICLR)*, 2026

Diffusion-based Adversarial Purification for Intrusion Detection

Mohamed Amine Merzouk, Erwan Beurier, Reda Yaich, Nora Boulahia-Cuppens, Frédéric Cuppens, Foutse Khomh

*Proceedings of the 39th Annual IFIP WG 11.3 Conference on Data and Applications Security and Privacy (DBSec)*, 2025

Parameterizing poisoning attacks in federated learning-based intrusion detection

Mohamed Amine Merzouk, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich

*Proceedings of the 18th International Conference on Availability, Reliability and Security (ARES)*, 2023

Evading Deep Reinforcement Learning-based Network Intrusion Detection with Adversarial Attacks

Mohamed Amine Merzouk, Joséphine Delas, Christopher Neal, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich

*Proceedings of the 17th International Conference on Availability, Reliability and Security (ARES)*, 2022

A Deeper Analysis of Adversarial Examples in Intrusion Detection

Mohamed Amine Merzouk, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich

*Proceedings of the 15th International Conference on Risks and Security of Internet and Systems (CRiSIS)*, 2021

### PREPRINTS

Quantifying Overclaiming Propensity in Frontier LLM Agents

Nolan Smyth, Yorguin-Jose Mantilla-Ramos, Pascal Jr Tikeng Notsawo, Saskia Helbling, Alberto Tosato, Mohamed Amine Merzouk, Nouha Dziri, Gauthier Gidel, Tommaso Tosato

Preprint, arXiv:2609.20812, 2026

Deep-Cover Agents: Long-Horizon Prompt Injections on Production LLM Systems

Neel Alex, Mohamed Amine Merzouk, David Krueger

Preprint, 2026

## THESES

A Landscape of Adversarial Threats to Machine Learning-Based Intrusion Detection

Mohamed Amine Merzouk

*Ph.D. Thesis. Polytechnique Montréal, 2024*

Adversarial Machine Learning against Anomaly-based Intrusion Detection Systems

Mohamed Amine Merzouk

*M.Sc. Thesis. École Nationale Supérieure d'Informatique (ESI ex. INI), 2020*

## AWARDS AND FUNDING

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|      |                                                                                                                 |                         |
|------|-----------------------------------------------------------------------------------------------------------------|-------------------------|
| 2025 | <b>AI Safety Catalyst Grant (with Pr. Khomh, Pr. Ma, and Pr. Goebel)</b> , Canadian AI Safety Institute & CIFAR | <i>Montréal, Canada</i> |
| 2024 | <b>Nomination for the Best Thesis Award</b> , Polytechnique Montréal                                            | <i>Montréal, Canada</i> |
| 2023 | <b>Best Presentation Award</b> , InCyber International Forum                                                    | <i>Montréal, Canada</i> |
| 2021 | <b>Mitacs Accelerate International Graduate Research Fellowship</b> , Polytechnique Montréal                    | <i>Montréal, Canada</i> |
| 2014 | <b>1st Place in Regional Mathematical Olympiad</b> , ASA Science                                                | <i>Algiers, Algeria</i> |

## INVITED TALKS

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|------|-----------------------------------------------------------------------------------------------------------|-------------------------|
| 2025 | <b>IRT SystemX</b> , Diffusion-based Adversarial Purification for Intrusion Detection                     | <i>Paris, France</i>    |
| 2023 | <b>InCyber International Forum</b> , Trustworthy AI (Best presentation award)                             | <i>Montréal, Canada</i> |
| 2023 | <b>BSides Algiers</b> , Adversarial threats at training and inference time                                | <i>Algiers, Algeria</i> |
| 2023 | <b>Cyberconference (Cybereco)</b> , Round table on AI security                                            | <i>Montréal, Canada</i> |
| 2022 | <b>School of AI Summit</b> , Adversarial Poisoning in Federated Learning                                  | <i>Algiers, Algeria</i> |
| 2022 | <b>International Cybersecurity Forum</b> , Adversarial Machine Learning Masterclass                       | <i>Montréal, Canada</i> |
| 2022 | <b>Canadian AI Federated Learning Workshop</b> , The Threat of Malicious Clients in Federated Learning    | <i>Toronto, Canada</i>  |
| 2022 | <b>Mila - Quebec Artificial Intelligence Institute</b> , Generating adversarial examples with constraints | <i>Montréal, Canada</i> |
| 2021 | <b>CRITiCAL chair launch</b> , Adversarial attacks against intrusion detection systems                    | <i>Paris, France</i>    |
| 2021 | <b>NeurIPS Meetup</b> , Adversarial attacks against intrusion detection systems                           | <i>Algiers, Algeria</i> |
| 2018 | <b>OpenStack User Group Meeting</b> , OpenStack Security                                                  | <i>Algiers, Algeria</i> |
| 2017 | <b>Docker Chapter</b> , Security Enhanced Linux                                                           | <i>Algiers, Algeria</i> |

## PEER REVIEW ACTIVITIES

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|                 |                                                                                   |
|-----------------|-----------------------------------------------------------------------------------|
| <b>IEEE</b>     | Transactions on Intelligent Vehicles                                              |
| <b>Elsevier</b> | Pattern Recognition                                                               |
| <b>Elsevier</b> | Computer Networks                                                                 |
| <b>Springer</b> | International Journal of Information Security                                     |
| <b>ESORICS</b>  | European Symposium on Research in Computer Security                               |
| <b>ISC</b>      | International Conference on Information Security                                  |
| <b>FPS</b>      | International Symposium on Foundations & Practice of Security                     |
| <b>WPES</b>     | Workshop on Privacy in the Electronic Society                                     |
| <b>SECRYPT</b>  | International Conference on Security and Cryptography                             |
| <b>CSNet</b>    | Cyber Security in Networking Conference                                           |
| <b>CASCON</b>   | IEEE International Conference on Collaborative Advances in Software and Computing |

## MENTORING

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**Ph.D. students** Mina Taraghi (graduated in 2026) and Masoud Karami, Polytechnique Montréal.

**M.Sc. students** Maxime Gourceyraud (graduated in 2024) and Joséphine Delas (graduated in 2023), Polytechnique Montréal.

**Undergraduates** Research interns Miran Qarachatani (Lakehead University), Adjobi Vernet Emmanuel Placide (Université de Moncton), Junyu Huo (University of Alberta) and Alfie Chuang (National Taiwan University).

# COMMUNITY AND SOCIAL CONTRIBUTIONS

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|----------|------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 2026-... | <b>Mentor</b> , SPAR (Supervised Program for Alignment Research)                                                       | Online                |
| 2026     | <b>Program committee</b> , Tool Demonstration Track, 17th International Conference on Internetware (Internetware 2026) | Gold Coast, Australia |
| 2026     | <b>Program committee</b> , CPRA workshop in conjunction with ARES 2026                                                 | Linköping, Sweden     |
| 2025-... | <b>Organizer</b> , Mila's AI Safety Reading Group                                                                      | Montréal, Canada      |
| 2025     | <b>Co-chair</b> , SEMLA Symposium 2025                                                                                 | Montréal, Canada      |
| 2025     | <b>Program committee</b> , CPRA workshop in conjunction with ARES 2025                                                 | Ghent, Belgium        |
| 2024-... | <b>Member</b> , Mila's Information Security Committee                                                                  | Montréal, Canada      |
| 2024-... | <b>Founder and organizer</b> , SEMLA Reading Group                                                                     | Montréal, Canada      |
| 2018-19  | <b>President</b> , Shellmates (first Algerian cybersecurity club)                                                      | Algiers, Algeria      |
| 2021-... | <b>Union representative</b> , Association pour une Solidarité Syndicale à l'École Polytechnique (ASSEP)                | Montréal, Canada      |
| 2023-... | <b>Animal rights activist</b> , WeTheFree                                                                              | Montréal, Canada      |
| 2020-21  | <b>Volunteer</b> , Secours Populaire Français                                                                          | Rennes, France        |
| 2021-24  | <b>Artistic mediator</b> , La Gang à Rambrou (community center for intellectual disability and autism)                 | Montréal, Canada      |
| 2021-... | <b>Theater actor</b> , Les Pains Perdus                                                                                | Montréal, Canada      |
| 2024-... | <b>Choir singer</b> , Canopées                                                                                         | Montréal, Canada      |
| 2023-... | <b>Guitarist</b> , The Tiny Tax Band                                                                                   | Montréal, Canada      |

# REFERENCES

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|                            |                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|
| <b>Adam Oberman</b>        | Postdoctoral advisor, Full Professor at McGill University and Academic Member at Mila |
| <b>Frédéric Cuppens</b>    | PhD supervisor, Full Professor at Polytechnique Montréal                              |
| <b>Reda Yaich</b>          | PhD co-supervisor, Head of the cybersecurity team at IRT SystemX                      |
| <b>Foutse Khomh</b>        | Full Professor at Polytechnique Montréal and Academic Member at Mila                  |
| <b>Yacine Challal</b>      | Graduate Program Director at ESI ex.INI                                               |
| <b>Fetheddine Guenaoui</b> | Senior Manager at KPMG                                                                |

Contact details are available on request.